

Single-Mode Dual-Band Patch Antenna Based on the Lorentz Model of Dispersive Metamaterials

Jiexi Yin¹, *Member, IEEE*, Zhi Ning Chen², *Fellow, IEEE*, Hanyang Wang³, *Fellow, IEEE*, Qun Lou⁴, *Member, IEEE*, and Ben Lai

Abstract—Based on the Lorentz model of dispersive metamaterials (Lorentz-dispersive metamaterials, in short), a method of generating single-mode dual-band operation of microwave patch antennas is presented. This method is inspired by the unique properties of Lorentz-dispersive materials, which exhibit a higher refractive index at lower frequencies and a lower refractive index at higher frequencies (relative to the material's resonant frequency). Based on the relation between these refractive indices and wavelengths of antennas, a patch antenna with a Lorentz-dispersive material substrate simultaneously operates at two distinct frequencies with a single electrical size. This phenomenon offers the possibility of generating single-mode radiation over two frequency bands for antenna designs. After studying the properties of the Lorentz-dispersive materials, a split-ring resonator (SRR) is simplified and miniaturized to realize a Lorentz-dispersive metamaterial. A single-mode dual-band patch antenna and its array with 1×4 elements are designed to verify the proposed method in sub-6-GHz bands. A rectangular patch antenna with the designed dispersive metamaterial realizes operation at two separate frequencies of 2.6 and 4.1 GHz simultaneously with the dominant TM_{10} mode. The antenna achieves nearly identical broadside radiation patterns over both frequency bands and gains of 6.9 and 8.1 dBi, respectively.

Index Terms—Dispersive metamaterial, dual-band antenna, Lorentz metamaterial, permeability, split-ring resonator (SRR).

I. INTRODUCTION

WIRELESS communication systems usually need dual-band (or even multiband) antennas to accommodate various applications [1]. A challenge facing these systems is that they usually require a dual-band antenna not only

for impedance matching but also for acceptable gains and consistent patterns over dual frequency bands, and they usually only occupy limited physical space [2], [3], [4], [5], [6].

A straightforward method for dual-band design is to combine or stack two antennas, each operating in a separate frequency band [7], [8], [9], [10]. However, this leads to a key issue where degraded performance is caused by the strong mutual coupling between the antennas, particularly due to the limited space they occupy. An alternative is the use of dual-band, aperture-sharing antennas, which are typically more compact [11], [12], [13], [14]. The aperture-sharing method can be realized by modifying the two operation modes, such as by using the TM_{10} and TM_{30} modes of patch antennas [11] and slot antennas [12]. A multimode design in one aperture is promising but challenging due to the difficulty in independently controlling the impedance and radiation performance of each mode. Recently, metasurface-based antennas (i.e., metantennas) have emerged as another solution for dual-band antenna design. Dual-band metantennas offer the advantages of low profile and broad bandwidth over conventional dual-band antennas. Dual-band metantennas can fulfill requirements, such as wide bandwidth [15], dual polarization [16], low profile [17], compact structure [18], and large frequency ratio [19], [20].

To date, most dual-band antennas have been designed using a multimode multiband method with at least two operating modes. However, considering impedance matching and consistent radiation patterns, a dual-band antenna operating in a single mode is desirable. A few dual-band antennas achieve dual-band operation in a single mode, such as stacked patch antennas, but these designs essentially require two structures of different physical sizes.

Metamaterials, or artificial electromagnetic (EM) structures, are capable of realizing permittivity and permeability properties that are not seen in natural materials. Both theoretical and applied research have shown that metamaterials offer opportunities to introduce new functions and enhance antenna performance [21], [22], [23], [24]. For example, in [25], the antenna is miniaturized by negative permittivity and permeability. Patch antennas integrated with zero-index metamaterials can achieve high directivity [26]. In [27], a printed antenna can achieve miniaturization and bandwidth improvement by placing U-shaped and inverted U-shaped metamaterials beneath the patch.

This article introduces a method based on Lorentz-dispersive metamaterials to realize single-mode dual-band

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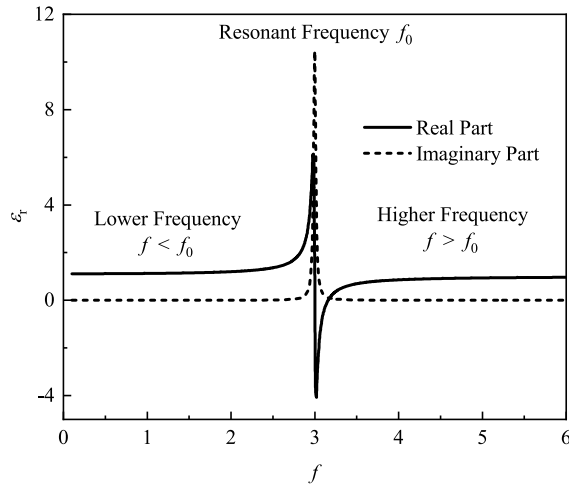


Fig. 1. Schematic curves of the Lorentz model ($\omega_p = 2\pi$, $\omega_0 = 6\pi$, and $\Gamma = 0.2$).

antenna designs. Lorentz-dispersive materials have unique properties that exhibit a higher refractive index at lower frequencies and a lower refractive index at higher frequencies (relative to the material's resonant frequency). Based on the relation between the refractive indices and wavelengths of antennas, one antenna is capable of operating at two distinct frequencies but with the same electrical size. Therefore, the antenna achieves single-mode dual-band operation. Lorentz permeability-dispersive materials are realized using simplified and miniaturized split-ring resonator (SRR) metamaterials. The metamaterial-based patch antenna is discussed in this article as an example, demonstrating how the proposed method can enable the dominant TM_{10} mode to operate in both designed bands.

The remainder of this article is structured as follows. Section II discusses the Lorentz-dispersive material and its potential applications in antenna design. Section III presents the unit cell with an SRR to realize the Lorentz permeability-dispersive metamaterial. Section IV details the design and verification of a patch antenna and a patch antenna array with 1×4 elements using the SRR metamaterial to achieve single-mode dual-band functionality. Both simulation and measurement results prove that using the proposed method, the SRR-based patch antenna can operate with the dominant TM_{10} mode at 2.6 and 4.1 GHz and can simultaneously achieve broadside radiation patterns.

II. THEORETICAL ANALYSIS OF THE LORENTZ-DISPERSIVE MATERIAL

A. Background of the Lorentz Model

The Lorentz model elucidates the dispersive characteristics of natural materials [28]. The properties of such materials, given the resonances of atoms and molecules, closely depend on the permittivity, $\tilde{\epsilon}_r$, especially around the resonant frequency. This permittivity can be derived as follows:

$$\tilde{\epsilon}_r(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - j\omega\Gamma} \quad (1)$$

where the angular frequency ω_p depends on the plasma frequency of the material. Items $\omega = 2\pi f$ and $\omega_0 = 2\pi f_0$ represent the angular frequency and angular resonant frequency, respectively. Γ is the damping coefficient. The parameters ω_p , ω_0 , and Γ are inherently dependent on the characteristics of the material.

The real and the imaginary parts of $\tilde{\epsilon}_r$ are

$$\text{Re}[\tilde{\epsilon}_r(\omega)] = 1 + \omega_p^2 \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \omega^2\Gamma^2} \quad (2)$$

and

$$\text{Im}[\tilde{\epsilon}_r(\omega)] = \omega_p^2 \frac{\omega\Gamma}{(\omega_0^2 - \omega^2)^2 + \omega^2\Gamma^2} \quad (3)$$

respectively. The loss tangent is

$$\tan \delta = \frac{\text{Im}[\tilde{\epsilon}_r(\omega)]}{\text{Re}[\tilde{\epsilon}_r(\omega)]}. \quad (4)$$

Fig. 1 shows the representative curves of the Lorentz model. Due to the resonance at ω_0 , the permittivity in the lower frequency (LF) band is greater than that in the higher frequency (HF) band. According to (4), the loss tangent of the material at ω_0 is

$$\tan \delta(\omega_0) = \frac{\omega_p^2}{\omega_0\Gamma}. \quad (5)$$

Fig. 2 shows a series of curves based on the Lorentz model with different parameters. It is found that ω_p and Γ influence the real part and imaginary part of $\tilde{\epsilon}_r$, especially near ω_0 . As ω_p increases, the slopes of $\text{Re}(\tilde{\epsilon}_r)$ and $\text{Im}(\tilde{\epsilon}_r)$ near ω_0 become steeper, and the difference in $\text{Re}(\tilde{\epsilon}_r)$ between the LF and the HF increases. Γ shapes the slopes of $\text{Re}(\tilde{\epsilon}_r)$ and $\text{Im}(\tilde{\epsilon}_r)$, as f is very close to f_0 . $\text{Re}(\tilde{\epsilon}_r)$ and $\text{Im}(\tilde{\epsilon}_r)$ further determine the loss of the material according to (4).

B. Analysis and Application of the Lorentz Model

When EM waves propagate through Lorentz-dispersive materials, they can retain their wavelengths λ at different frequencies f . This intriguing behavior is explained by considering the relationship λ for the relative permittivity ϵ_r and permeability μ_r

$$\lambda(f) = \frac{c_0}{f\sqrt{\epsilon_r\mu_r}} \quad (6)$$

or refractive index n

$$\lambda(f) = \frac{c_0}{fn} \quad (7)$$

where c_0 is the speed of light in free space.

Assuming two frequencies f_1 and f_2 with $f_1 < f_2$ and according to (6) and (7), $\lambda(f_1) = \lambda(f_2)$ if

$$f_1\sqrt{\epsilon_r(f_1)\mu_r(f_1)} = f_2\sqrt{\epsilon_r(f_2)\mu_r(f_2)} \quad (8)$$

or

$$f_1n(f_1) = f_2n(f_2). \quad (9)$$

Fig. 1 illustrates that n can satisfy this condition when f_1 is below the resonant frequency f_0 , and f_2 is above f_0 . This is attributed to $\epsilon_r(f_1)$ being larger than $\epsilon_r(f_2)$. Moreover, for

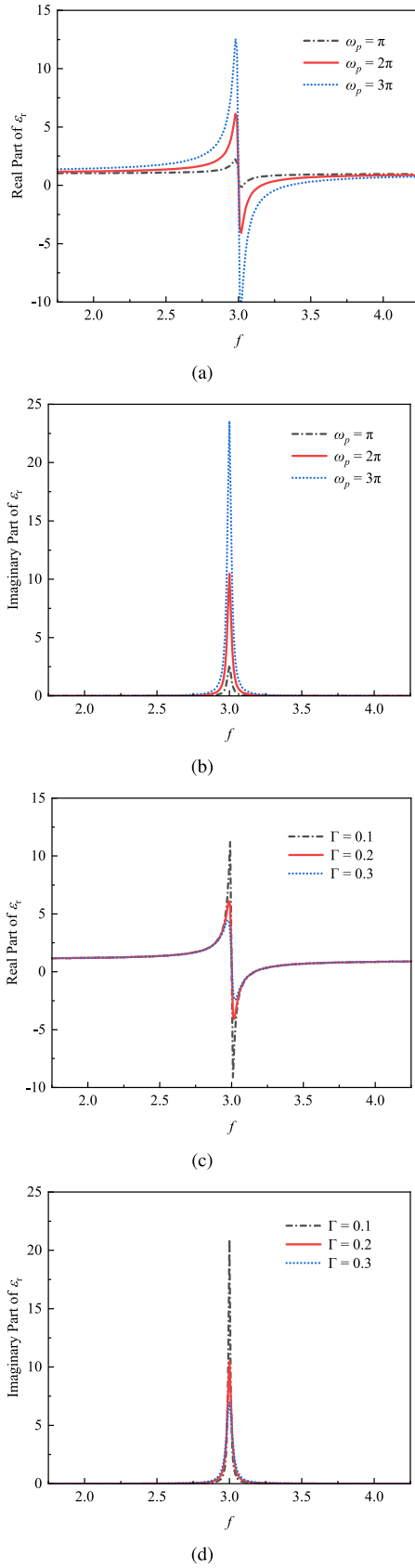


Fig. 2. Curves of the real parts and imaginary parts with different values of ω_p and Γ . (a) Real part and (b) imaginary part of $\tilde{\epsilon}_r$ with different values of ω_p and $\Gamma = 0.2$. (c) Real part and (d) imaginary part of $\tilde{\epsilon}_r$ with different values of Γ and $\omega_p = 2\pi$.

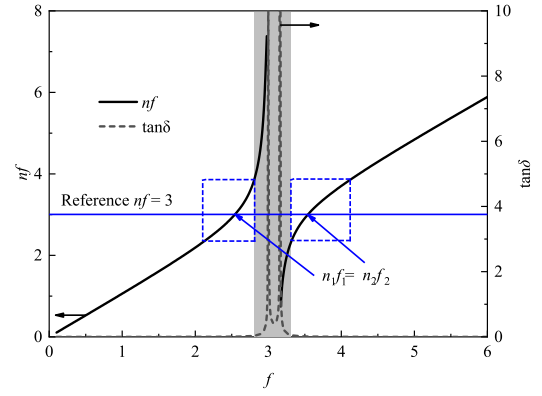


Fig. 3. Schematic curves of the Lorentz model ($\omega_p = 2\pi$, $\omega_0 = 6\pi$, and $\Gamma = 0.2$). Relationship between f and nf for the Lorentz model.

any chosen f_1 that is less than f_0 , there must be an associated f_2 above f_0 that satisfies (9).

This analysis illustrates our new method for designing single-mode dual-band antennas. If a reference antenna is constructed using a Lorentz-model-dispersive material and its operation mode is established at f_1 , then the antenna can also operate at f_2 in the same operation mode. In contrast with the reference antenna deployed on a standard dielectric substrate, an additional operation frequency can be attained. This feature allows a single-mode antenna to support dual-band operation, which is a significant advancement in antenna design strategy. Furthermore, considering an antenna loaded with Lorentz-dispersive material operating at N distinct modes, it is theoretically feasible for our antenna to operate at twice as many (i.e., $2N$) frequencies for the same N operation modes. Thus, Lorentz-dispersive materials potentially hold the ability to double the operating frequency range without increasing the complexity of operation modes.

For antenna designs, operation frequencies and efficiency are of greater interest than $Re(\tilde{\epsilon}_r)$ and $Im(\tilde{\epsilon}_r)$. The slopes of the Lorentz curves significantly influence f_1 and f_2 in antenna design. Fig. 3 exhibits the relationship between frequency f and nf for the Lorentz model with unit μ_r . According to (9), the intersections between the blue line and black curve indicate that the wavelengths at corresponding frequencies are the same. This signifies that for any chosen f_1 ranging from 1 to 3, there exists a solution for f_2 ranging from 3 to 6, where both frequencies exhibit the same electrical size.

However, near the resonance frequency, $\tan \delta$ is considerable. This indicates that if the antenna operates within the gray area of Fig. 3, its efficiency will drastically decrease. Therefore, the antenna operation bands should be out of the resonant frequency range of the material.

The frequency ranges enclosed by the blue dotted lines indicate where the material has a low loss tangent $\tan \delta$, and for any f_1 , there is a solution of f_2 in (9). These ranges can be considered the effective bandwidth of the material for single-mode dual-band antenna designs. Fig. 3 also reveals that the available bandwidths of the material and solutions of f_1 and f_2 are determined by the given Lorentz model and $\tan \delta$.

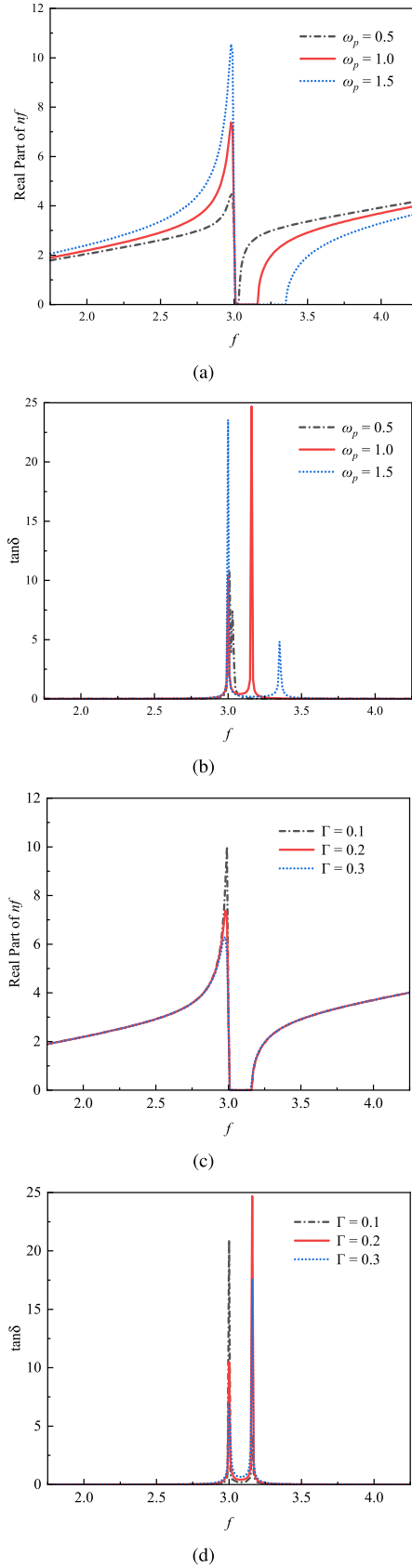


Fig. 4. Curves of the real parts of nf and loss of the material with different values of ω_p and Γ . (a) Real parts of nf with different values of ω_p and $\Gamma = 0.2$. (b) Loss of the material with different values of ω_p and $\Gamma = 0.2$. (c) Real parts of nf with different values of Γ and $\omega_p = \pi$. (d) Loss of the material with different values of Γ and $\omega_p = \pi$.

Fig. 4 presents a parametric analysis for ω_p and Γ . ω_p is related to f_2/f_1 and the loss of material. For large ω_p , f_2/f_1 is large, while the unavailable bandwidth with large $\tan \delta$ near f_0 is increased. Γ significantly influences the material loss near f_0 ; however, $\tilde{\epsilon}_r$ and nf are nearly unchanged for different Γ , as f is far from the resonant frequency f_0 . $\tan \delta$ reveals that the material incurs high loss not only at the resonant frequency but also at the frequencies where the real part of $\tilde{\epsilon}_r$ is nearly zero.

C. Analysis of Patch Antenna Based on Lorentz Model

A rectangular patch antenna is presented in Fig. 5, and it typically operates in the dominant TM_{10} mode. At a frequency f , the length of the patch, L , for the TM_{10} mode can be approximated as follows:

$$L \approx \frac{c_0}{2f\sqrt{\epsilon_r\mu_r}}. \quad (10)$$

If the refractive index n of the substrate satisfies (9), the patch antenna can operate in the TM_{10} mode at f_1 and f_2 . The patch dimensions, L and W , can be calculated based on the desired frequencies and μ_r and ϵ_r of the substrate. As discussed, the Lorentz model displayed in Fig. 1 can provide a higher refractive index at f_1 and a lower refractive index at f_2 if $f_1 < f_0 < f_2$. Therefore, Lorentz-dispersive materials could theoretically be utilized for single-mode dual-band patch antenna design.

Characteristic mode analysis (CMA) elucidates the physical characteristics of the patch antenna loaded with Lorentz-dispersive materials. The dispersive material is set as the Lorentz model with $\omega_p = 3 \times 2\pi \times 10^9$ rad/s, $\Gamma = 0.2$ s⁻¹, and $\epsilon_r = 1.0$. However, only $Re(\tilde{\epsilon}_r)$ is considered in the CMA. The patch, with $L = 36$ mm and $W = 45$ mm, as shown in Fig. 5, is mounted above the Lorentz-dispersive material without a feed, and the simulated model significances (MSs) of different modes are exhibited in Fig. 6. The case in the gray area is omitted because of the high material loss at this frequency band.

Fig. 6 demonstrates that the TM_{01} mode includes two operation frequencies of 2.35 and 3.28 GHz, while the TM_{10} mode includes two operation frequencies of 2.73 and 3.63 GHz. The refractive indices and the frequencies roughly satisfy the conditions in (9). Fig. 7 reveals that the TM_{01} mode provides similar electric (E)-field distributions at the two frequencies. These simulated results confirm the theoretical analysis, underlining the potential of Lorentz-dispersive materials in single-mode dual-band antenna designs.

III. REALIZATION OF PERMEABILITY-DISPERSIVE METAMATERIAL AT MICROWAVE BANDS

The above analysis is based on the mathematical Lorentz model. However, f_0 , ω_p , and Γ essentially depend on the materials. For natural materials, f_0 often appears above infrared or even ultraviolet frequency ranges due to the high resonant frequency of the atoms in natural materials. Few engineering applications exploit such dispersive dielectric materials near f_0 , especially in the context of antenna designs. Furthermore,

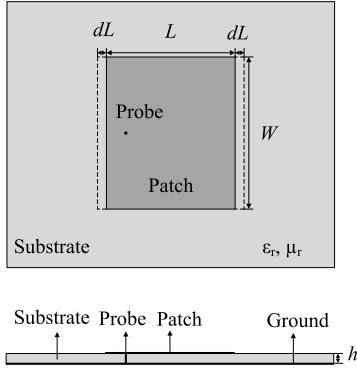


Fig. 5. Model of a conventional patch antenna.

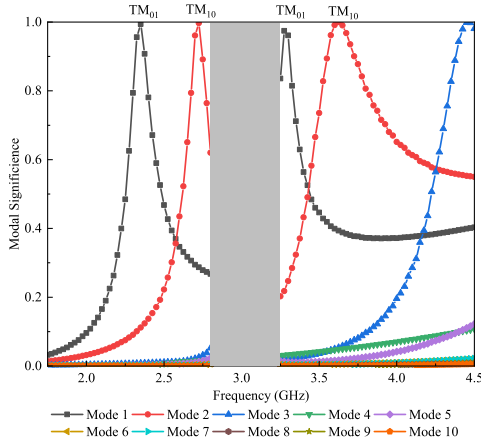


Fig. 6. CMA for the patch antenna with the Lorentz-dispersive material as the substrate.

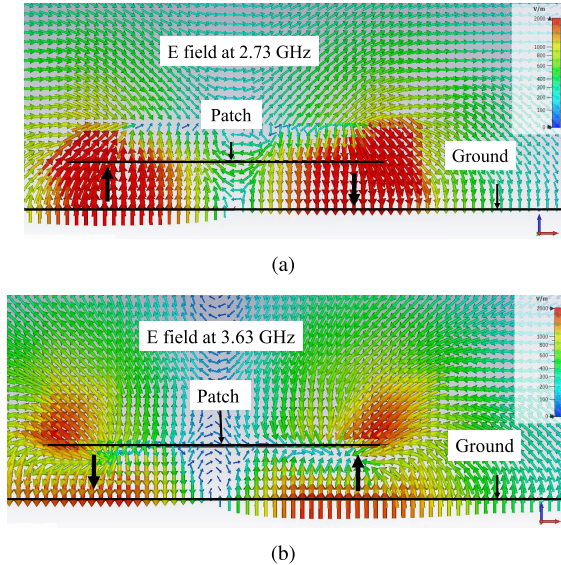


Fig. 7. E-field distributions of the patch antenna at (a) 2.73 and (b) 3.63 GHz.

only permittivity-dispersive Lorentz materials are observed in natural materials, as outlined in Fig. 1, primarily because no magnetic monopole has been discovered thus far.

Metamaterials, which are artificially designed structures, mimic the unit cell behavior seen in natural materials, where

the unit cell is akin to that of atoms. This allows the realization of a dispersive curve similar to the Lorentz model. Metamaterials control both permeability and permittivity, aligning with the Lorentz model through the intentional design of unit cells. For instance, an electric resonance unit cell, such as a strip line structure, can be leveraged for the design of dispersive permittivity [29], while a magnetic resonance unit cell, such as a split ring structure, can be employed for permeability-dispersive design [30]. Moreover, any resonant frequency can be tailored by manipulating the electrical length of the strip line or the perimeter of the split rings.

Similar to (1), the relative effective dispersive permeability of the metamaterial, denoted as $\tilde{\mu}_r$, using the Lorentz model, is derived as follows:

$$\tilde{\mu}_r(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - j\omega\Gamma}. \quad (11)$$

In this case, ω_p , ω_0 , and Γ are determined by the unit cell of the metamaterial.

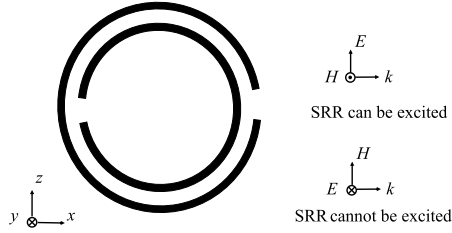
The SRR provides a feasible approach for realizing metamaterial design with Lorentz-dispersive permeability. Furthermore, it is typically easier to miniaturize the unit cells by manipulating permeability compared with permittivity. Hence, for the purpose of single-mode dual-band antenna design, a modified version of the conventional SRR is employed.

A. Mechanism of SRR

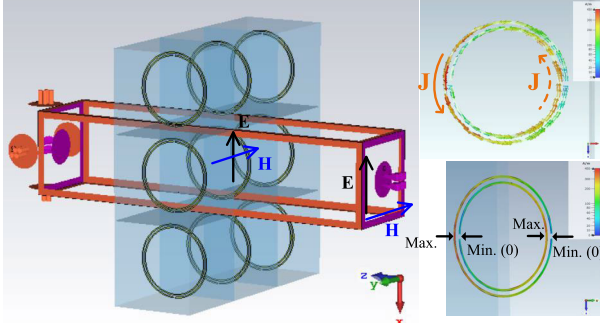
As shown in Fig. 8, the SRR consists of two split rings. These rings create complementary currents, where a maximum current in one ring compensates for the minimum current at the split slots of the other. This design allows the SRR to mimic a loop current, which is akin to a magnetic dipole. While the use of SRRs in antenna designs has been explored in previous works, such as [31], [32], and [33], the utilization of SRRs as a dispersive metamaterial for antenna design has not been exploited yet.

Similar to atomic and molecular resonances for dispersive permittivity, the resonance of SRR results in Lorentz-dispersive permeability. Physically, an arbitrary resonant frequency f_0 can be achieved by tailoring the geometrical parameters of the SRR, i.e., the length of the current path. However, we note that the SRR structure, as shown in Fig. 8, is anisotropic. Only magnetic fields that pass through the loop area and generate significant magnetic flux can effectively couple with the SRR. According to the analysis in [30], the Lorentz-dispersive curve can be observed only when the SRR is excited. Otherwise, the metamaterial behaves as a nondispersive substrate with constant permeability and permittivity. For example, in Fig. 8(a), the SRR is excited when the magnetic (H)-field is perpendicular to the area enclosed by the SRR but not when the H-field is parallel to this area. Therefore, the metamaterial design should align with the field distributions of the antenna.

The effective parameters of the unit cell, μ_r and ϵ_r , can be calculated as the method outlined in [30]. Here, the parameters μ_r and ϵ_r are extracted with commercial EM software, CST. As shown in Fig. 8(b), an SRR is placed in the unit cell boundary, with two ports providing specific E- and H-fields.

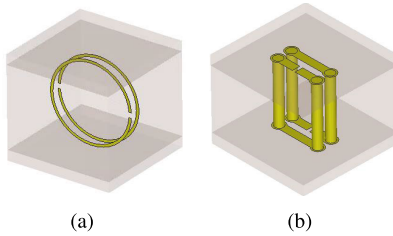


(a)



(b)

Fig. 8. Model of a conventional SRR [30]. (a) Conventional SRR and (b) SRR in the unit cell boundary and the current on the rings at the SRR resonant frequency.



(a)

(b)

Fig. 9. Improvement of the conventional SRR for the antenna design. (a) Circular SRR. (b) Rectangular SRR.

Then, the effective parameters of the unit cell can be extracted through the S-parameters of these two ports.

B. Modified SRR

The traditional SRR can be modified and miniaturized for metamaterial design, as depicted in Fig. 9. Fig. 9(a) presents the conventional microstructure for magnetic resonators [25]. However, this design poses challenges for printed circuit board (PCB)-based fabrication and integration with patch antennas. Drawing inspiration from the structure in [29], the conventional SRR can be modified into the structure shown in Fig. 9(b), which is constructed using metallic vias and strips that are suitable for the PCB process, thus facilitating easy integration of SRRs with patch-type antennas.

To preserve the dispersive property, the unit cell's size should be significantly smaller than the antenna; otherwise, SRR unit cells cannot function as an equivalent dispersive medium [30]. Considering the size and thickness of the patch antenna, the SRR shown in Fig. 9(b) is further modified,

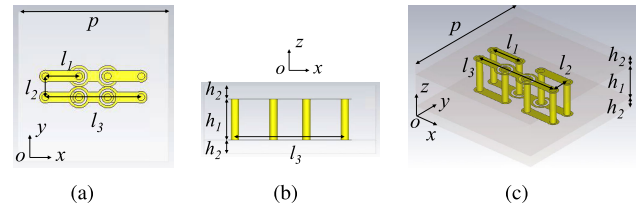


Fig. 10. Final SRR model for the antenna design—(a) top view, (b) side view, and (c) 3-D view ($l_1 = 2.8$ mm, $l_2 = 1.75$ mm, $l_3 = 8.0$ mm, $h_1 = 3.0$ mm, $h_2 = 1.0$ mm, and $p = 12.5$ mm.)

as illustrated in Fig. 10. By integrating additional meanders, the final version of the SRR not only meets the requirements for PCB fabrication but also satisfies the miniaturization and low profile. This structure can be implemented by single-layer substrate, double metal layers, and vias.

The SRR was initially designed based on [30] and then optimized by CST. The F4BM220 substrate with $\epsilon_r = 2.2$ is used. The overall size of the unit cell is approximately $0.12 \times 0.12 \times 0.05 \lambda_0^3$ (λ_0 denotes the wavelength at the resonant frequency of 3.1 GHz). The unit cell exhibits an anisotropic EM response and dispersive property. When the SRR is excited, the permeability of the metamaterial fits the Lorentz model.

C. Parametric Analysis of SRR

The SRR unit cell is simulated in the unit cell boundary with the H-field perpendicular to the plane of SRR, as shown in Fig. 8(a). The simulated μ_r and n are presented in Fig. 11. High values of μ_r and n can be achieved near the resonant frequency of the SRR. Beyond the resonant frequency, n increases from 0. Similar to the Lorentz model, n also increases from 0 for atom resonance, and n can be large before the resonant frequency and very small after, ensuring any f_1 in (9) has a corresponding f_2 solution, resulting in a single mode for dual bands. Moreover, the material loss decreases rapidly before and after the resonant frequency, ensuring the radiation efficiency of the antenna. The metamaterial is suitable for single-mode dual-band antenna designs when the metamaterial loss remains low. In the LF band, for frequencies less than 1.5 GHz, the metamaterial behaves as a nondispersive material with μ_r close to the unit.

The most critical parameter for the SRR is the length of the ring, which determines the resonant frequency f_0 of the unit cell. Various parameters of the unit cells affect n , as depicted in Fig. 12. l_3 controls the length of the ring, thereby significantly influencing the resonant frequency. The spacing between two rings, l_2 , also influences the resonant frequency to a certain extent. The period (p) of the unit cell influences the ratio of f_1 and f_2 .

IV. SINGLE-MODE DUAL-BAND PATCH ANTENNA DESIGN

The structure of the single-mode dual-band patch antenna is presented in Fig. 13. Similar to the unit cell design, F4BM220 with $\epsilon_r = 2.2$ is used as the substrate for the metamaterial and antenna. A microstrip line and a slot on the ground serve as the feeding network to avoid additional feeding structures between

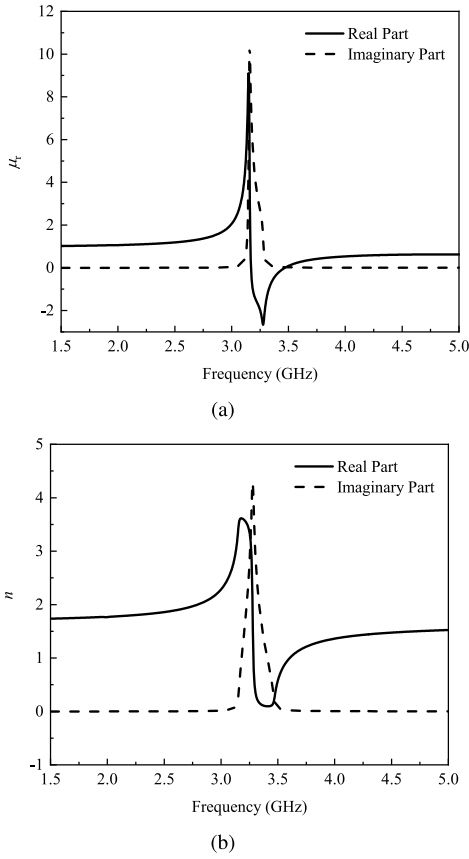


Fig. 11. Simulated results of the designed SRR—(a) equivalent permeability (μ_r) and (b) equivalent refractive index (n) of the designed unit cell.

the patch and its ground. The entire antenna comprises four single-layer substrates, as illustrated in Fig. 13(a). Substrate 1 includes a patch; Substrate 2 is a double metal layer SRR-based metamaterial with metalized holes; Substrate 3 is only a dielectric board without a metal layer, which is utilized to separate SRRs and ground; and Substrate 4 includes double metal layers, one for the ground and the other for the feed line. The ground of Substrate 4 is shared by the antenna and the feed line. This feed line excites the patch antenna through a slot on the ground. All layers are assembled with nylon screws, which are inserted away from both the antenna and SRRs to reduce their influence on the operating performance.

The SRR unit cell depicted in Fig. 10 is used for the dispersive metamaterial design. The SRRs are arranged according to the H-field distribution of the TM_{10} mode, and the metamaterial with SRRs is placed under the patch structure. The metamaterial only needs to fill the cavity of the patch. Based on the effective refractive index, the physical size of the antenna can be estimated according to the designed frequencies, which then allows for the calculation of the number of SRRs. An array of 2×4 SRRs, as shown in Fig. 10, is used. The arrangement of the SRRs is presented in Fig. 13(a) with the prototype depicted in Fig. 13(b). Due to the limited number of SRRs, the effective refractive index differs from the unit cell boundary condition in SRR unit cell simulation. Therefore, the parameters of the antenna and the SRR are further optimized for the desired frequencies and impedance matching. Parameters for optimization can include the size of

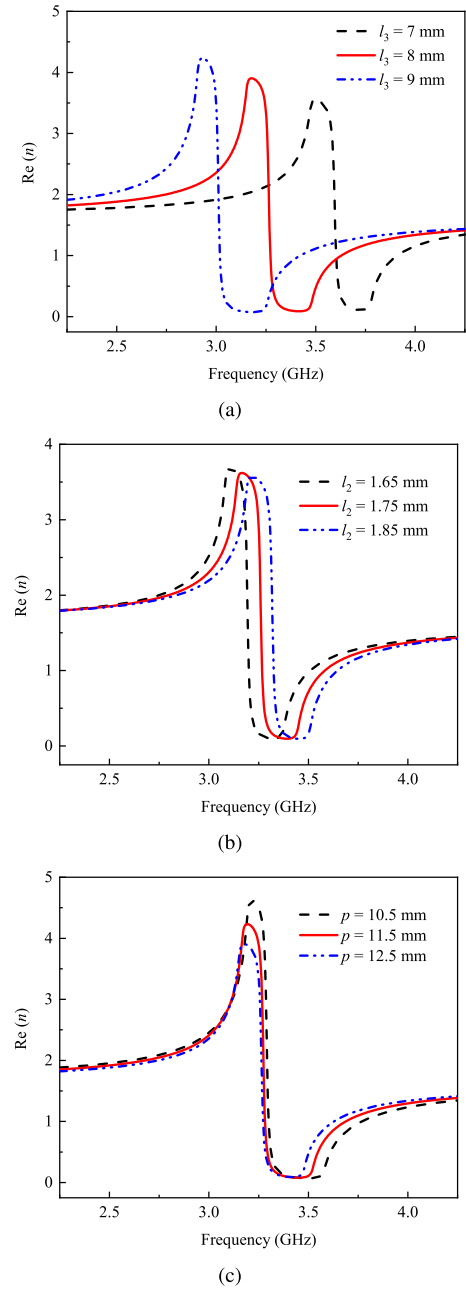


Fig. 12. Parametric analysis of SRR. (a) n with different lengths (l_3) of the rings. (b) n with different distances (l_2) of two rings. (c) n with different periods (p) of the unit cell.

the patch, L_2 and W_2 , the length of the slot W_3 , the length L_4 and width W_4 of the feeding network, the period of the unit cell p , and the length of the SRR, l_1 – l_3 .

The Southwest end launch connector is utilized to connect the antenna for testing. In this particular work, the specific type of connector utilized is the End Launch 2.92-mm (K) 40-GHz Jack (Female) Standard Block [34]. The simulated $|S_{11}|$ and gain of the patch antenna are presented in Fig. 14. For comparison, a patch antenna without SRR is simulated, and the simulation results are also presented in Fig. 14. The size of the patch and metamaterial-based patch, L_2 and W_2 , and the thickness of the substrate, H_1 – H_4 , are kept the same. The permittivity of the substrate of the compared patch antenna

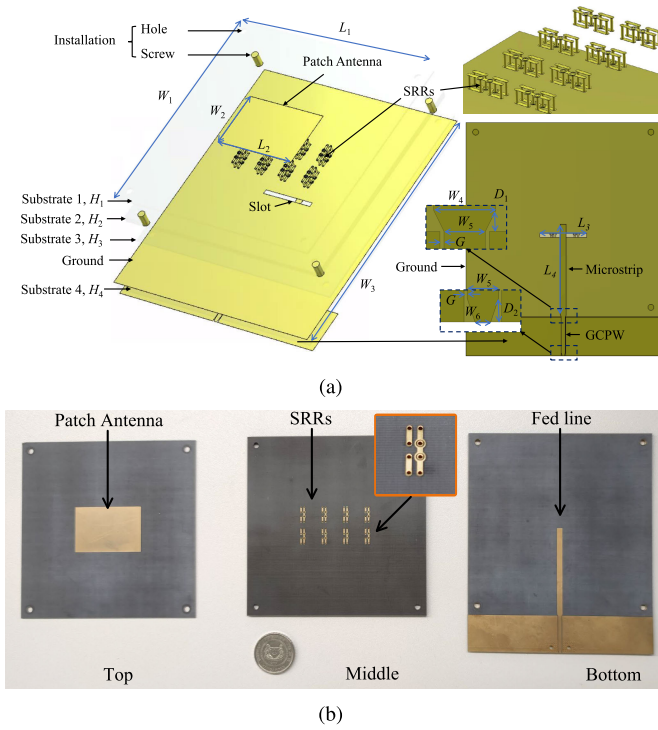


Fig. 13. Structure of the single-mode dual-band patch antenna. (a) 3-D model of the proposed antenna and (b) photograph of the proposed antenna: top view and bottom view ($L_1 = 100$ mm, $L_2 = 37.5$ mm, $L_3 = 24.0$ mm, $L_4 = 46.75$ mm, $W_1 = 100$ mm, $W_2 = 25.5$ mm, $W_3 = 120$ mm, $W_4 = 3.0$ mm, $W_5 = 2.0$ mm, $W_6 = 1.0$ mm, $H_1 = 1.0$ mm, $H_2 = 3.0$ mm, $H_3 = 1.0$ mm, $H_4 = 1.0$ mm, $D_1 = 1.0$ mm, $D_2 = 1.5$ mm, and $G = 0.2$ mm).

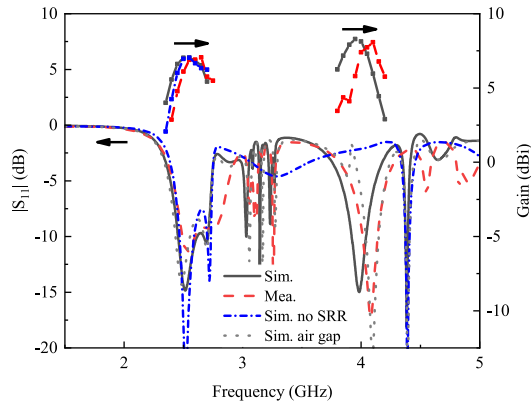


Fig. 14. Simulated and measured $|S_{11}|$ and gain of the patch antennas.

(without SRR) is set as 3.0 to tune the operation frequency of the TM_{10} mode near 2.5 GHz.

The simulation results in Fig. 14 show that the impedance bandwidths of the conventional antenna and the metamaterial-based patch antenna are 4.7% (2.47–2.59 GHz) and 6.3% (2.46–2.62 GHz), respectively. The simulated gains of the two antennas are both approximately 7.0 dBi near 2.5 GHz. However, due to the dispersive SRR structure, the metamaterial-based antenna has an additional operation bandwidth with the same mode, dominant TM_{10} mode, at 4.0 GHz. The simulated gain at 4.0 GHz is 8.3 dBi. The simulated radiation efficiency of the conventional antenna is 99% at

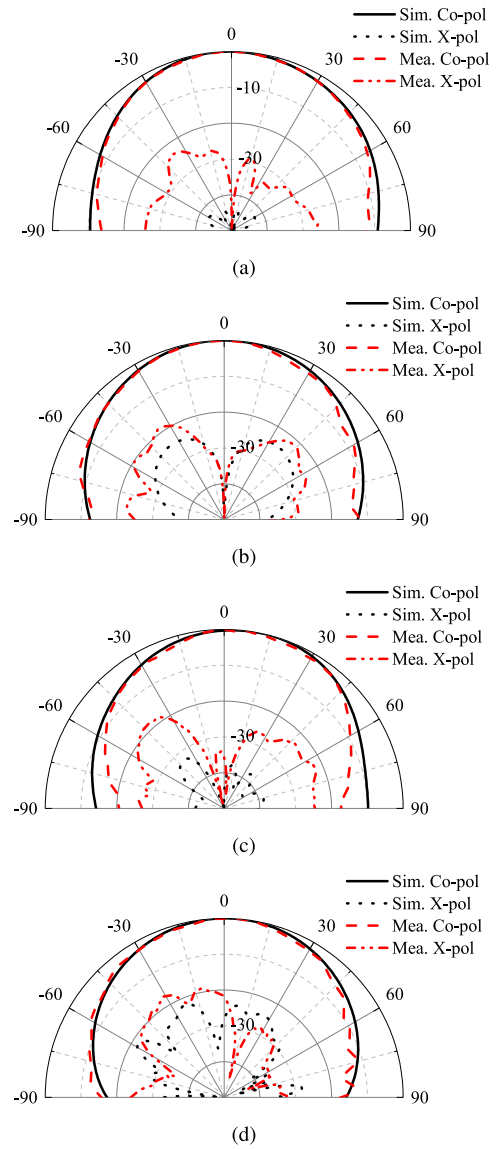


Fig. 15. Simulated and measured radiation patterns of the patch antenna in (a) E-plane at 2.6 GHz, (b) H-plane at 2.6 GHz, (c) E-plane at 4.0 GHz, and (d) H-plane at 4.0 GHz.

2.5 GHz, and the SRR-based antenna is 98% and 95% at 2.5 and 4.0 GHz, respectively.

The measured bandwidth of $|S_{11}| < -10$ dB is 3.9% (2.50–2.60 GHz) and 2.9% (4.02–4.14 GHz) for dual bands, respectively. This bandwidth depends on the effective ϵ_r and μ_r of the metamaterial and the thickness of the patch. The measured frequencies are 2.6 and 4.1 GHz, with a gain of 6.9 and 8.1 dBi, respectively.

In the LF band, another pseudoresonance near 2.7 GHz is introduced by the feeding network. However, this pseudoresonance does not map onto the HF band, as the H-field in the feeding network does not interact with the SRR. Due to the air gaps between layers and the tolerance of PCB fabrication, the measured frequencies are found to be slightly higher than the simulated results. As presented in Fig. 14, the simulated result incorporating an air gap of 0.15 mm between substrates reveals that the operating frequencies shift to higher frequencies when

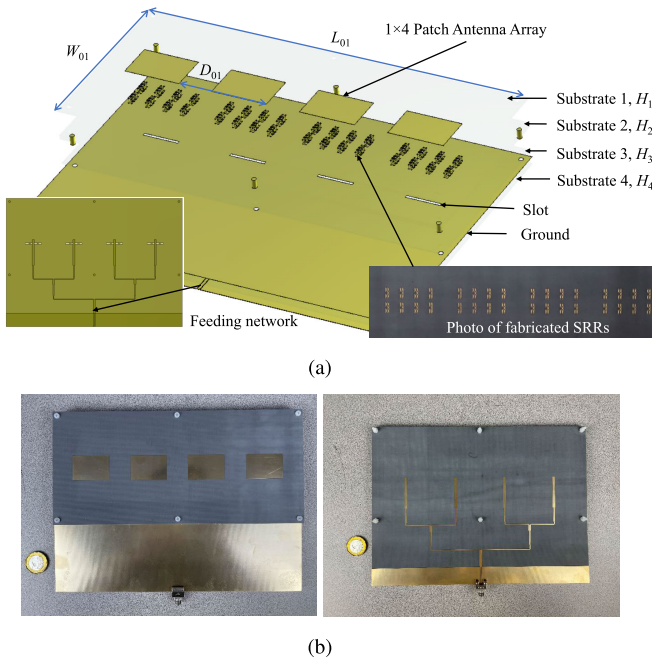


Fig. 16. Structure of the single-mode dual-band 1×4 patch antenna array. (a) 3-D model of the proposed antenna and (b) photograph of the proposed antenna: top view and bottom view ($L_{01} = 260$ mm, $W_{01} = 100$ mm, and $D_{01} = 60$ mm).

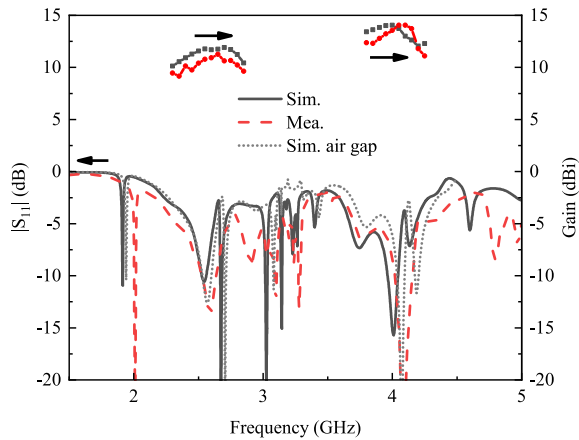


Fig. 17. Simulated and measured $|S_{11}|$ and gain of the 1×4 patch antenna.

air gaps exist. The simulated and measured radiation patterns at 2.5 and 4.0 GHz are presented in Fig. 15, demonstrating the antenna's stable broadside radiation at both the LF and HF bands.

Table I compares the proposed metamaterial-based dual-band patch antenna with conventional multimode dual-band patch antennas. All compared antennas can realize broadside patterns in dual bands. Compared with these published works, the key advantage of the proposed method lies in its ability to generate consistent patterns over two bands while maintaining a relatively small size and very high radiation efficiency at a price of additional SRRs between the patch and the ground.

A linear antenna array with 1×4 antenna elements is further developed to demonstrate the feasibility of our patch antenna element for array design with high gain, as depicted in Fig. 16. To provide a uniform phase and amplitude for

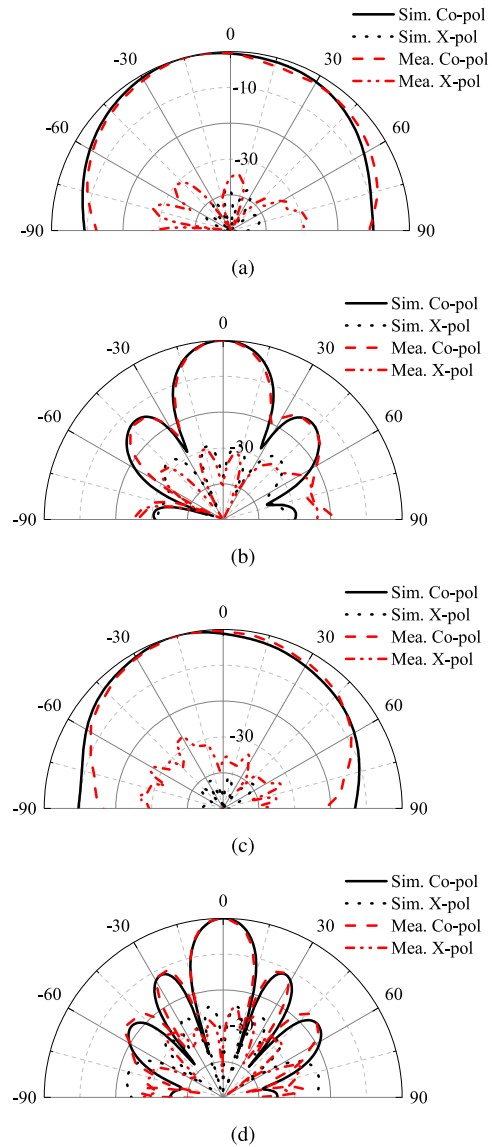


Fig. 18. Simulated and measured radiation patterns of the patch antenna array in (a) E-plane at 2.6 GHz, (b) H-plane at 2.6 GHz, (c) E-plane at 4.0 GHz, and (d) H-plane at 4.0 GHz.

each element, a 1-to-4 power divider is designed with a microstrip line structure. The antenna array achieved high gain and broadside radiation at two distinct frequencies. The simulated $|S_{11}|$ and gain of the patch antenna array are shown in Fig. 17. Pairs of simulated and measured frequencies are at 2.5 and 4.0 GHz, and 2.6 and 4.1 GHz, corresponding to the gains of 11.6 and 14.0 dBi, and 10.9 and 14.1 dBi, respectively. The measured bandwidth of $|S_{11}| < -10$ dB is 3.9% (2.53–2.63 GHz) and 2.9% (4.03–4.15 GHz) for dual bands, respectively. Simulated results considering an air gap of 0.2 mm between substrates are also shown in Fig. 17. Similar to the behavior of the antenna element, the presence of an air gap serves to shift the operating frequencies to higher values. The bending of the substrate also influences $|S_{11}|$ to a certain degree. The bandwidth of the array is similar to the element. Fig. 18 illustrates the simulated and measured patterns, which agree well. The designed antenna element and array may have

TABLE I
COMPARISONS OF THE PROPOSED SINGLE-MODE DUAL-BAND ANTENNA WITH THOSE DUAL-BAND PATCH ANTENNAS IN REFERENCES

Ref.	Freq. (GHz)	Bandwidth (%)	Gain (dBi)	ϵ_r	Method	Size (λ)	Efficiency (%)*
[11]	0.61/1.87	2/8	N.G.	2.6	TM ₀₁ , TM ₀₃	0.40×0.30×0.007	N.G.
[17]	3.6/5.8	1.4/0.8	6.4/5.3	3.0	TM ₀₁ , TM ₁₀	0.26×0.26×0.02	N.G.
[24]	2.4/5.2	1.9/3.6	2.6/6.6	3.0	TM ₀₁	0.23×0.17×0.012	N.G.
[13]	3.8/5.2	5.2/9.6	9.0/9.1	1.0	Two U-slots	0.45×0.33×0.076	N.G.
[35]	3.61/5.90	9/5	6.5/8.0	2.2	TM ₁₀ , TM ₁₂	0.55×0.84×0.048	90
[36]	1.9/3.5	4/2.2	1/2.4	4.4	Pin-loaded modes	0.30×0.18×0.11	28/35
[37]	1.95/2.45	4.5/5	8.3/7.8	3.55, 4.4, 1.06	Stacked disc	0.43×0.43×0.11	90
This work	2.55/4.11	3.92/2.94	7.0/8.3	2.2	TM ₀₁	0.32×0.22×0.043	98/95

* N.G. represents “Not Given”; The efficiency in [38] is radiation efficiency; The efficiency in this work is simulated radiation efficiency.

potential applications for 5G Wi-Fi and cellular frequency bands.

Compared with the theoretical study of the Lorentz model and CMA data for the dispersive metamaterial-based patch antenna, the simulated and measured results of the antenna element and array verify the practical feasibility of the proposed single-mode dual-band method in antenna designs. By employing Lorentz-dispersive metamaterials, the typical patch antenna has been successfully excited at an additional frequency in the same mode, such as a TM₁₀ mode, resulting in similar radiation patterns. Slight variations in the gains and beam widths in the LF and HF bands are observed, attributable to the differing effective electrical sizes in free space.

V. CONCLUSION

Lorentz-dispersive metamaterials have been presented for single-mode dual-band antenna designs. Guided by the Lorentz model, the designed metamaterial was demonstrated to exhibit a higher refractive index at frequencies below the resonant frequency and a lower refractive index at frequencies above the resonant frequency. This unique property of the Lorentz-dispersive metamaterial has enabled realization of a single antenna that can concurrently operate at two frequencies while maintaining the same mode, a feature denoted as single-mode dual-band operation.

For the realization of the antenna design, a simplified SRR has been presented and acts as a Lorentz-dispersive metamaterial. The SRR was fabricated using the PCB process at sub-6-GHz bands. Using the proposed SRR structure, a rectangular patch antenna operating in the TM₁₀ mode is designed. The study has revealed that the proposed dispersive metamaterial-based antenna can simultaneously operate at two distinct frequencies in the TM₁₀ mode. The study further indicated that metamaterials with dispersive permittivity or permeability have offered the feasibility of dual-band antenna designs, potentially leading to further advances in antenna technology.

REFERENCES

- [1] T. S. Rappaport et al., “Millimeter wave mobile communications for 5G cellular: It will work!” *IEEE Access*, vol. 1, pp. 335–349, 2013.
- [2] T. Zhang, W. Hong, Y. Zhang, and K. Wu, “Design and analysis of SIW cavity backed dual-band antennas with a dual-mode triangular-ring slot,” *IEEE Trans. Antennas Propag.*, vol. 62, no. 10, pp. 5007–5016, Oct. 2014.
- [3] S. X. Ta and I. Park, “Dual-band low-profile crossed asymmetric dipole antenna on dual-band AMC surface,” *IEEE Antennas Wireless Propag. Lett.*, vol. 13, pp. 587–590, 2014.
- [4] T. Deckmyn, M. Cauwe, D. Vande Ginste, H. Rogier, and S. Agneessens, “Dual-band (28,38) GHz coupled quarter-mode substrate-integrated waveguide antenna array for next-generation wireless systems,” *IEEE Trans. Antennas Propag.*, vol. 67, no. 4, pp. 2405–2412, Apr. 2019.
- [5] A. T. Mobashsher, M. T. Islam, and N. Misran, “A novel high-gain dual-band antenna for RFID reader applications,” *IEEE Antennas Wireless Propag. Lett.*, vol. 9, pp. 653–656, 2010.
- [6] S. Li, Z. Yi, and G. Yang, “Single-layer antenna with dual-band for 5G millimeter-wave system,” in *Proc. Int. Symp. Antennas Propag. (ISAP)*, Jan. 2021, pp. 391–392.
- [7] T. Li, H. Meng, and W. Dou, “Design and implementation of dual-frequency dual-polarization slotted waveguide antenna array for Ka-band application,” *IEEE Antennas Wireless Propag. Lett.*, vol. 13, pp. 1317–1320, 2014.
- [8] P. Li, K. M. Luk, and K. L. Lau, “A dual-feed dual-band L-probe patch antenna,” *IEEE Trans. Antennas Propag.*, vol. 53, no. 7, pp. 2321–2323, Jul. 2005.
- [9] K. Geary, J. H. Schaffner, H.-P. Hsu, H. J. Song, J. S. Colburn, and E. Yasan, “Single-feed dual-band stacked patch antenna for orthogonal circularly polarized GPS and SDARS applications,” in *Proc. IEEE 68th Veh. Technol. Conf.*, Sep. 2008, pp. 1–5.
- [10] H. Huang, Y. Liu, and S. Gong, “A novel dual-broadband and dual-polarized antenna for 2G/3G/LTE base stations,” *IEEE Trans. Antennas Propag.*, vol. 64, no. 9, pp. 4113–4118, Sep. 2016.
- [11] S. S. Zhong and Y. T. Lo, “Single-element rectangular microstrip antenna for dual-frequency operation,” *Electron. Lett.*, vol. 19, no. 8, pp. 298–300, Apr. 1983.
- [12] W.-J. Lu and L. Zhu, “Wideband stub-loaded slotline antennas under multi-mode resonance operation,” *IEEE Trans. Antennas Propag.*, vol. 63, no. 2, pp. 818–823, Feb. 2015.
- [13] W. C. Mok, S. H. Wong, K. M. Luk, and K. F. Lee, “Single-layer single-patch dual-band and triple-band patch antennas,” *IEEE Trans. Antennas Propag.*, vol. 61, no. 8, pp. 4341–4344, Aug. 2013.
- [14] M. Veysi, M. Kamyab, and A. Jafarholi, “Single-feed dual-band dual-linearly-polarized proximity-coupled patch antenna,” *IEEE Antennas Propag. Mag.*, vol. 53, no. 1, pp. 90–96, Feb. 2011.
- [15] T. Li and Z. N. Chen, “A dual-band metasurface antenna using characteristic mode analysis,” *IEEE Trans. Antennas Propag.*, vol. 66, no. 10, pp. 5620–5624, Oct. 2018.
- [16] B. P. Smyth and A. K. Iyer, “Embedded MTM-EBGs in patch antenna for simultaneously dual-band and dual-polarized operation,” in *Proc. 14th Eur. Conf. Antennas Propag. (EuCAP)*, Mar. 2020, pp. 1–4.

- [17] B. P. Smyth and A. K. Iyer, "Low-profile uniplanar dual-band and dual-polarized microstrip patch antenna using embedded MTM-EBGs," *IEEE Trans. Antennas Propag.*, vol. 69, no. 7, pp. 3645–3653, Jul. 2021.
- [18] E. Baladi, M. Sempale, and A. K. Iyer, "Single-layer dual-band polarization-selective metafilm with independently controlled and closely spaced shielding bands," *IEEE Trans. Antennas Propag.*, vol. 68, no. 3, pp. 1448–1457, Mar. 2020.
- [19] T. Li and Z. N. Chen, "Shared-surface dual-band antenna for 5G applications," *IEEE Trans. Antennas Propag.*, vol. 68, no. 2, pp. 1128–1133, Feb. 2020.
- [20] T. Li and Z. N. Chen, "Metasurface-based shared-aperture 5G S-/K-band antenna using characteristic mode analysis," *IEEE Trans. Antennas Propag.*, vol. 66, no. 12, pp. 6742–6750, Dec. 2018.
- [21] C. Miliat et al., "Metamaterial-inspired antennas: A review of the state of the art and future design challenges," *IEEE Access*, vol. 9, pp. 89846–89865, 2021.
- [22] Y. Dong and T. Itoh, "Metamaterial-based antennas," *Proc. IEEE*, vol. 100, no. 7, pp. 2271–2285, Jul. 2012.
- [23] Z. N. Chen, Q. Lou, and W. Liu, "Metantennas: Opportunities and challenges in future microwave metasurface antenna research and applications," in *Proc. 16th Eur. Conf. Antennas Propag. (EuCAP)*, Mar. 2022, pp. 1–5.
- [24] B. P. Smyth, S. Barth, and A. K. Iyer, "Dual-band microstrip patch antenna using integrated uniplanar metamaterial-based EBGs," *IEEE Trans. Antennas Propag.*, vol. 64, no. 12, pp. 5046–5053, Dec. 2016.
- [25] S. F. Mahmoud, "A new miniaturized annular ring patch resonator partially loaded by a metamaterial ring with negative permeability and permittivity," *IEEE Antennas Wireless Propag. Lett.*, vol. 3, pp. 19–22, 2004.
- [26] H. Zhou et al., "A novel high-directivity microstrip patch antenna based on zero-index metamaterial," *IEEE Antennas Wireless Propag. Lett.*, vol. 8, pp. 538–541, 2009.
- [27] S. A. H. Saghanézhad and Z. Atlasbaf, "Miniaturized dual-band CPW-fed antennas loaded with U-shaped metamaterials," *IEEE Antennas Wireless Propag. Lett.*, vol. 14, pp. 658–661, 2015.
- [28] H. A. Lorentz and H. Lorentz, *Versuch Einer Theorie Der Elektrischen Und Optischen Erscheinungen In Bewegten Körpern*. Springer, 1937. [Online]. Available: https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=Versuch+einer+Theorie+der+elektrischen+und+optischen+Erscheinungen+in+bewegten+K%C3%B6rpern&btnG=
- [29] D. Schurig, J. J. Mock, and D. R. Smith, "Electric-field-coupled resonators for negative permittivity metamaterials," *Appl. Phys. Lett.*, vol. 88, no. 4, Jan. 2006, Art. no. 041109.
- [30] J. B. Pendry, A. J. Holden, D. J. Robbins, and W. J. Stewart, "Magnetism from conductors and enhanced nonlinear phenomena," *IEEE Trans. Microw. Theory Techn.*, vol. 47, no. 11, pp. 2075–2084, Nov. 1999.
- [31] F. J. Herraiz-Martinez, L. E. Garcia-Munoz, D. Gonzalez-Ovejero, V. Gonzalez-Posadas, and D. Segovia-Vargas, "Dual-frequency printed dipole loaded with split ring resonators," *IEEE Antennas Wireless Propag. Lett.*, vol. 8, pp. 137–140, 2009.
- [32] W. Cao, B. Zhang, A. Liu, T. Yu, D. Guo, and Y. Wei, "Gain enhancement for broadband periodic endfire antenna by using split-ring resonator structures," *IEEE Trans. Antennas Propag.*, vol. 60, no. 7, pp. 3513–3516, Jul. 2012.
- [33] M. Hu, Z. Yu, J. Xu, J. Lan, J. Zhou, and W. Hong, "Diverse SRRs loaded millimeter-wave SIW antipodal linearly tapered slot filter with improved stopband," *IEEE Trans. Antennas Propag.*, vol. 69, no. 12, pp. 8902–8907, Dec. 2021.
- [34] 1092-03A-6 End Launch 2.92 mm (K) 40 GHz Jack Female Standard Block. Accessed: Aug. 1, 2023. [Online]. Available: <https://mpd.southwestmicrowave.com/product/1092-03a-6-end-launch-2-92mm-k-40-ghz-jack-female-standard-block/>
- [35] N.-W. Liu, L. Zhu, Z.-X. Liu, and Y. Liu, "Dual-band single-layer microstrip patch antenna with enhanced bandwidth and beamwidth based on reshaped multiresonant modes," *IEEE Trans. Antennas Propag.*, vol. 67, no. 11, pp. 7127–7132, Nov. 2019.
- [36] A. Jafarholi, A. Jafarholi, and B. Ghalamkari, "Dual-band slim microstrip patch antennas," *IEEE Trans. Antennas Propag.*, vol. 66, no. 12, pp. 6818–6825, Dec. 2018.
- [37] M. Aboulalaa, A. B. Abdel-Rahman, A. Allam, H. Elsadek, and R. K. Pokharel, "Design of a dual-band microstrip antenna with enhanced gain for energy harvesting applications," *IEEE Antennas Wireless Propag. Lett.*, vol. 16, pp. 1622–1626, 2017.



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